Teaching notes and solutions

Teacher note · Teacher notes. The scheme of work condenses this chapter into three sessions: genuinely split S29 between differentiation and convexity, then revisit the proof about tangents in S33. The vocabulary continuous/differentiable/convex must be distinguished by examples. Continuity gives existence via the IVT; strict monotonicity gives uniqueness. For bisection, require an initial interval where the sign changes and a continuous function. A change of convexity, and not merely the vanishing of f'', characterises a point of inflection.

Solutions to the activity

1. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ and $f''(x) = 6x$. The sign of f' is positive on $]-\infty, -1[$ and $]1, +\infty[$, zero at $-1, 1$, negative between these values. The sign of f'' is negative to the left of 0, zero at 0, positive to the right.

2. The variations and the limits are summarised by

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

   The sign of f'' gives concavity on $]-\infty, 0]$ and convexity on $[0, +\infty[$.

3. The convexity changes at 0, and $f(0) = 0$: the point of inflection is $(0, 0)$. Its tangent has equation $y = f(0) + f'(0)x = -3x$.

4. The function is continuous. On $]-\infty, -1]$, it is strictly increasing from $-\infty$ to 2: the level 1 is reached exactly once. On $[-1, 1]$, it strictly decreases from 2 to -2: a second solution. On $[1, +\infty[$, it strictly increases from -2 to $+\infty$: a third. The points $-1, 1$ are not solutions, so no solution is counted twice.

5. Continuity provides the existence of a preimage for intermediate values. It does not provide uniqueness: the function studied is continuous and the level 1 has three preimages.

E1. For $u = 2x - 1$, $(u^5)' = 5u'u^4 = 10(2x-1)^4$. For $v = -x^2$, $(e^v)' = v'e^v = -2xe^{-x^2}$. Both functions are differentiable on $\mathbb{R}$.

E2. $f' = 4x^3 - 12x$ then $f'' = 12(x^2 - 1)$. The sign of f'' is positive outside $[-1, 1]$, negative inside. So f is convex on $]-\infty, -1]$ and $[1, +\infty[$, concave on $[-1, 1]$. The changes of convexity give the points of inflection $(-1, -5)$ and $(1, -5)$.

E3. The tangent is $y = f(1) + f'(1)(x-1) = e + e(x-1) = ex$. Since $f''(x) = e^x > 0$, the exponential is convex and its curve lies above this tangent: $e^x \geq ex$ on $\mathbb{R}$.

E4. Set $g(x) = x^3 + 2x - 5$. This polynomial is continuous and $g'(x) = 3x^2 + 2 > 0$ on $\mathbb{R}$. Moreover $g(1) = -2 < 0$ and $g(2) = 7 > 0$: the IVT gives a zero in $]1, 2[$, and strict increase on $\mathbb{R}$ ensures that it is unique.

E5. After n steps, the length is $\frac{2}{2^n}$. The required condition is equivalent to $2^n \geq 2000$. Since $2^{10} = 1024 < 2000 \leq 2048 = 2^{11}$, eleven steps suffice and ten do not.

Solutions to Homework 4

1. $f(0) = 1$ and $\lim_{x\to 0^+} f(x) = 1$. At $+\infty$, $f(x) = xe^{-x} + e^{-x} \to 0$ by comparison of growth rates. By the product rule, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. This derivative is strictly negative for $x > 0$: f strictly decreases from 1 towards 0 on $[0, +\infty[$.

2. $f''(x) = -e^{-x} + xe^{-x} = (x-1)e^{-x}$. So f is concave on $[0,1]$, convex on $[1, +\infty[$. The change of convexity at 1 gives the point of inflection $(1, \frac{2}{e})$.

3. f is continuous and strictly decreasing on $[0, +\infty[$; $f(0) = 1 > \frac{1}{2}$ and its limit 0 is less than $\frac{1}{2}$. The corollary of the IVT gives a unique solution α. Moreover, $f(1) = \frac{2}{e} \simeq 0.7358 > \frac{1}{2}$ and $f(2) = \frac{3}{e^2} \simeq 0.4060 < \frac{1}{2}$, so $1 < \alpha < 2$.

4. ```
   from math import exp

   def racine_dm4():
       a, b = 0.0, 2.0
       while b-a > 1e-3:
           m = (a+b)/2
           if (m+1)*exp(-m) > 0.5:
               a = m
           else:
               b = m
       return a, b
   ```

   The decrease preserves $f(a) \geq \frac{1}{2} \geq f(b)$; α stays in $[a, b]$. After eleven steps, we obtain $[1.677734375, 1.6787109375]$, of length $\frac{1}{1024} < 10^{-3}$. The value $\alpha \simeq 1.67835$ is consistent with this bracketing.

5. The tangent at 0 is $y = f(0) + f'(0)x = 1$. The variations give $f(x) \leq 1$, so the whole curve lies below this tangent. This does not characterise concavity: the curve would have to lie below all the tangents, whereas f becomes convex after 1.
   ```

Solutions to the textbook or past-paper work

For $f(x) = 5x^2e^x$, the product rule gives

$$f'(x) = 10xe^x + 5x^2e^x = 5x(x+2)e^x.$$

Since $e^x > 0$, f' is positive on $(-\infty, -2)$ and $(0, +\infty)$, negative on $(-2, 0)$, zero at -2 and 0. The function therefore increases up to $f(-2) = 20e^{-2}$, decreases down to $f(0) = 0$, then increases. Its limits are 0 at $-\infty$ (comparison of growth rates) and $+\infty$ at $+\infty$. Differentiating again gives

$$f''(x) = 5(x^2 + 4x + 2)e^x = 5\big((x+2)^2 - 2\big)e^x.$$

Set $a = -2 - \sqrt{2}$ and $b = -2 + \sqrt{2}$. The polynomial factor is positive outside $[a, b]$, negative in (a, b). Thus f is convex on $(-\infty, a]$ and $[b, +\infty)$, concave on $[a, b]$. The sign changes at the two roots: the points of inflection are $(a, 5a^2e^a)$ and $(b, 5b^2e^b)$.

The table of variations is summarised as follows:

x	$-\infty$		-2		0		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	0	$\nearrow$	$20e^{-2}$	$\searrow$	0	$\nearrow$	$+\infty$

For question 5, the window below makes the two changes of convexity and the two extrema visible. The marked points correspond to the points of inflection calculated; the plot serves to check the consistency of the study.

